ANAGHA ANEESH

PhD Student | Chemistry x AI

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RESEARCH EXPERIENCE

PhD Student

Stanford University Sept 2025 – Current

- Researcher in **Prof. Grant Rotskoff's** group, training and evaluating LLMs for structural chemistry, reaction prediction, and synthesis planning
- Run continued pre-training and GRPO post-training experiments of chemical reaction prediction data on small, open-weight models
- Designed and built an end-to-end research codebase covering tokenization, SFT, RL post-training, and benchmarking, with Hydra-based configuration for full reproducibility
- Built chemistry-aware dataset pipelines (PubChem PUG REST scraping, BitBIRCH clustering on Morgan fingerprints, functional-group-balanced train/test splits) and custom benchmarks for structural chemistry tasks

Fulbright Research Student

Friedrich Schiller University Jena Sept 2024 – July 2025

- Full-time researcher in **Prof. Kevin Jablonka's** AI for chemistry and materials group
- Built an ML pipeline predicting device-level properties of perovskite solar cells from compositional and processing descriptors, handling leakage from chemically correlated features in multi-source experimental data

Undergraduate Research Student

Haverford College May 2022 – May 2024

- Researcher in **Prof. Clyde Daly's** computational chemistry group
- Developed a predictive spectroscopic map for terminal alkyne vibrational probes by combining DFT (QChem), ALMO-EDA charge-transfer analysis, classical molecular dynamics, and ML regression on electrostatic features
- Senior thesis and first-author manuscript published

Undergraduate Researcher *(Total Synthesis CURE)*

Haverford College Sept 2022 – May 2023

- Year-long research course in **Prof. Grant Spoors'** group on green chemistry approaches to pharmaceutical synthesis
- Redesigned the synthetic route to an endothelin antagonist to replace lithium-mediated steps with magnesium-mediated Grignard chemistry; validated a solvent change to acetonitrile that improved yield

PUBLICATIONS

Journal Articles

- **A. Aneesh**, N. Alampara, M. Ríos-García, *et al.*, "General-purpose models for the chemical sciences: Llms and beyond," *Chemical Reviews*, vol. 126, no. 4, pp. 2484–2549, 2026, PMID: 41642594. DOI: 10.1021/acs.chemrev.5c00583. eprint: https://doi.org/10.1021/acs.chemrev.5c00583.
- **A. Aneesh**, K. Streu, and C. A. Daly Jr., "Development of a spectroscopic map to explain the broad raman peak for alkynes solvated in triethylamine," *Journal of Physical Chemistry B*, 2025.

SUMMARY

First-year Chemistry PhD student at Stanford working at the intersection of **large language models and small-molecule chemistry**. Trained as both a synthetic and computational chemist, with hands-on experience training and evaluating LLMs for chemical reasoning, reaction prediction, and synthesis planning. Long-term goal: AI systems that can propose synthetic routes to **novel molecules** from scratch.

TECHNICAL SKILLS

LLM Training & Post-Training

HuggingFace Transformers, TRL, PEFT, vLLM, DeepSpeed ZeRO, GRPO, supervised fine-tuning, continued pre-training, lm-evaluation-harness

Cheminformatics & Computational Chemistry

RDKit, SMARTS / reaction SMARTS, Morgan fingerprints, DFT (QChem), ALMO-EDA, classical molecular dynamics, PubChem PUG REST

ML & Data

PyTorch, scikit-learn, NumPy, pandas, Matplotlib, Seaborn

Infrastructure

SLURM, multi-node H100 / distributed training, Hydra, Weights & Biases, Bash, Git, Linux / HPC environments

Languages

Python (proficient), JavaScript / HTML / CSS (basic)

AWARDS

- NSF Graduate Research Fellowship, 2025
- Fulbright Germany Research Award, 2024
- ACS Undergraduate Physical Chemistry Award, 2024
- Liberal Arts in the Workplace Grant, Haverford College, 2022

EDUCATION

PhD | Chemistry

Stanford University Sept 2025 – Current

BS | Chemistry and Physics

Haverford College Sept 2020 – May 2024

- A. Mirza, N. Alampara, S. Kunchapu, *et al.*, “A framework for evaluating the chemical knowledge and reasoning abilities of large language models against the expertise of chemists,” *Nature Chemistry*, vol. 17, pp. 1027–1034, 2025. DOI: 10.1038/s41557-025-01815-x.

Conference Proceedings

- **A. Aneesh**, N. Alampara, J. A. Márquez, and K. M. Jablonka, “Semantic device graphs for perovskite solar cell design,” in *International Conference on Learning Representations*, Published as a workshop paper at AI4MAT at ICLR 2025, ICLR, 2025.
- **A. Aneesh**, K. Streu, and C. A. Daly Jr., “Explaining terminal alkyne solvatochromism with flexible vibrational quantum mechanics,” in *American Chemical Society Fall Conference*, Accepted abstract and poster, ACS, 2023.

RELEVANT COURSEWORK

- CS229: Introduction to Machine Learning, Stanford (Autumn 2025)
- CS109: Introduction to Algorithms and Data Structures, Haverford (Fall 2023)